

Problem 1

Solution:

Lemma: if $n > 1$ is a positive integer and p is a positive real, $0 < \frac{p}{n} < p$.

Proof: Since n is positive, $n^{-1} > 0$. We can multiply through $n > 1$ by this n^{-1} to give $n^{-1}n = 1 > 1(n^{-1}) = n^{-1}$. Now, multiplying through again (by p this time) produces $1(p) = p > n^{-1}p$. Also, $n^{-1}p > 0$ since n^{-1} and p are both positive. We can rewrite this as $0 < \frac{p}{n} < p$.

The proof of the main assertion follows directly from the lemma. By substituting $p = (y - x)$, which we know is > 0 since $x < y$, and $n = 2$, we get $0 < \frac{(y-x)}{2} < (y - x)$. This is equivalent to $x < x + \frac{(y-x)}{2} < y$. Hence, for any real $x < y$, $z = x + \frac{(y-x)}{2}$ lies between x and y . $\square$

Problem 2

Solution:

By theorem 1.29, there exists a positive integer n such that $x < n$. And since $-x$ is also real, there exists a positive integer k such that $-x < k$. But this implies that $x > -k$. If we let $m = -k$, we see that $m < x < n$, as desired. $\square$

Problem 3

Solution:

By the Archimedean property, there exists a positive integer n such that $nx > 1$. Which, when multiplied through by n^{-1} produces $x > \frac{1}{n}$

$\square$

Problem 4

Solution:

We will prove the statement by contradiction. Suppose that there is not exactly one integer n that satisfies $n \leq x < n + 1$. We have two cases: there are no such n or there are at least two such n .

We will start with the former. If there are no such n that satisfy $n \leq x < n + 1$, then every integer n either satisfies $x < n$ or $x \geq n + 1$. Now, consider the set $S = \{s \in \mathbb{Z} : s + 1 \leq x\}$. Observe that S has a maximum element because otherwise $s < s + 1 \leq x$ (i.e. x is an upper bound for the set of all integers), which is a contradiction by theorem 1.28.

Let $b = \max(S)$. Then, b is an integer where $b + 1 \leq x$ (since $b \in S$) and $b = \sup(S)$. Now, we have 2 cases. If $b + 1 = x$, then we have a contradiction – there exists an integer n where $n + 1 = x + 1 > x$ and $n = x \not\geq x$. And if $b + 1 < x$, we have a contradiction because $b + 1 \in S \wedge b + 1 > b$, i.e. b isn't an upper bound for S (and therefore not a supremum).

Now, the second case. If there are at least two integers p, q that satisfy the inequality, then we have $p \leq x < p + 1$ and $q \leq x < q + 1$. The second can be rewritten as $-q - 1 < -x \leq -q$. And then we have $(p - q) - 1 < 0 \leq (p - q) + 1$, which simplifies to $-1 < q - p \leq 1$. There are now two sub-cases: $q - p = 0$ or $q - p = 1$.

If $q - p = 1$ then $p = q - 1$ and we arrive at both $x < q$ (by substituting for p in $x < p + 1$) and $x \geq q$ (directly from $q \leq x < q + 1$). But this is a contradiction since only one can be true (due to the trichotomy property).

On the other hand, if $q - p = 0$, then $q = p$. But this too is a contradiction since we assumed that there are two distinct integers that satisfy the inequality. We are now done since we have illustrated contradictions in all of the cases.

□

Problem 5

Solution:

From problem 4, we know that for arbitrary real x , $[x] \leq x < [x] + 1$, where $[x]$ is the greatest integer in x . Therefore, there are only two possible cases: $[x] = x$ and $[x] < x$. In the first case, observe that $x = [x] < [x] + 1 = x + 1$, so $n = [x]$ satisfies the inequality we're trying to prove. In the second case, $[x] < x$ implies $[x] + 1 < x + 1$, which, when combined with the $x < [x] + 1$ part of problem 4, gives us $x < [x] + 1 < x + 1$. That is, $n = [x] + 1$ satisfies the inequality we're trying to show.

Now we will show uniqueness. Suppose $n_1 \leq x < n_1 + 1$ and $n_2 \leq x < n_2 + 1$ where $n_1 \neq n_2$. Then, $-n_2 - 1 < -x \leq -n_2$ so we have $n_1 - n_2 - 1 < 0 < n_1 - n_2 + 1$. But this implies $-1 < n_2 - n_1 < 1$, which means $n_2 - n_1 = 0$, or rather $n_1 = n_2$. Contradiction!

□

Problem 6

Solution:

Since $x < y$, we know $y - x > 0$. Therefore, by the Archimedean property, there exists an integer n such that $n(y - x) > 1$. That is, $1 < ny - nx$. Now, by exercise 5 we also know that there exists a unique integer m such that $nx \leq m < nx + 1$. Hence, $x \leq \frac{m}{n}$ and $m < nx + (ny - nx) = ny \implies \frac{m}{n} < y$. We can combine these into $x \leq \frac{m}{n} < y$. Note that $n \neq 0$ because otherwise $n(y - x) = 0(y - x) = 0 > 1$ – a contradiction.

Now, there are two cases. If $x < \frac{m}{n} < y$, then we are done right away, i.e. we have found a rational r such that $x < r < y$. However, if $x = \frac{m}{n}$, then we have $1 < ny - nx = ny - m$. But this means, $\frac{m+1}{n} < y$. Thus, we have $x = \frac{m}{n} < \frac{m+1}{n} < y$. But since m, n are integers and $n \neq 0$, $\frac{m+1}{n}$ is rational too. Hence, even in this case there exists a rational r such that $x < r < y$.

Now, observe that we can repeat the procedure over and over, i.e. we can find a rational r_1 such that $x < r_1 < \frac{m}{n} < y$, a rational r_2 such that $x < r_2 < r_1 < y$, and so on. This gives us a sequence of rationals $r_1, r_2, \dots$ and from theorem 1.28, we know that there is no upper bound for the set of positive integers. Thus, the existence of one rational between arbitrary reals implies the existence of infinitely many.

□

Problem 7

Solution:

We will show $x + y \in \mathbb{R} - \mathbb{Q}$ by contradiction. Suppose that $z = x + y$ is rational. Then, $y = z - x$. Let $z = \frac{a}{b}$, $b \neq 0$ and $x = \frac{c}{d}$, $d \neq 0$. From problem 10 in section 3, we know that $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$. And by the contrapositive of theorem 1.11, since neither b nor d is 0, $bd \neq 0$. Therefore y is rational, which is a contradiction.

We will show $x - y \in \mathbb{R} - \mathbb{Q}$ by contradiction. Suppose that $z = x - y$ is rational. Then, $y = x - z$. Let $x = \frac{a}{b}$, $b \neq 0$ and $z = \frac{c}{d}$, $d \neq 0$. We can use the exact same logic from above (omitted for brevity) to show that y is rational. Contradiction.

We will show $\frac{x}{y} \in \mathbb{R} - \mathbb{Q}$ by contradiction. Suppose that $z = \frac{x}{y}$ is rational. Then, $y = \frac{x}{z}$. Let $x = \frac{a}{b}$, $b \neq 0$ and $z = \frac{c}{d}$, $d \neq 0$. By theorem 1.15, we know that $\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{ad}{bc}$. Now, we must show that $bc \neq 0$. To do so, observe that if $c = 0$, then $z = \frac{c}{d} = \frac{0}{d} = 0$ and if $z = 0$, then $z = 0 = \frac{x}{y} \implies x = 0$. However, $x \neq 0$ by assumption. Therefore, we have $b \neq 0$ and $c \neq 0$, which implies, by the contrapositive of theorem 1.11, that $bc \neq 0$. Thus y is rational, which is a contradiction.

We will show $\frac{y}{x} \in \mathbb{R} - \mathbb{Q}$ by contradiction. Suppose that $z = \frac{y}{x}$ is rational. Then, $y = xz$. Let $x = \frac{a}{b}$, $b \neq 0$ and $z = \frac{c}{d}$, $d \neq 0$. By theorem 1.14, we know that $\frac{a}{b} \left(\frac{c}{d} \right) = \frac{ac}{bd}$. And by the contrapositive of theorem 1.11, since neither b nor d is 0, $bd \neq 0$. Therefore y is rational, which is a contradiction.

We will show $xy \in \mathbb{R} - \mathbb{Q}$ by contradiction. Suppose that $z = xy$ is rational. Then, $y = \frac{z}{x}$. Let $x = \frac{a}{b}$, $b \neq 0$ and $z = \frac{c}{d}$, $d \neq 0$. By theorem 1.15, we know that $\frac{\frac{c}{d}}{\frac{a}{b}} = \frac{ad}{bc}$. Now, we must show that $bc \neq 0$. To do so, observe that if $c = 0$, then $z = \frac{c}{d} = \frac{0}{d} = 0$ and if $z = 0$, then $y = \frac{0}{x} = 0$. However, that would mean that y is rational, which is a contradiction. Therefore, we have $b \neq 0$ and $c \neq 0$, which implies, by the contrapositive of theorem 1.11, that $bc \neq 0$. Thus y is rational anyway, which is a contradiction.

□

Problem 8

Solution:

The sum of two irrational numbers may or may not be irrational. For example $1 + \sqrt{2} - \sqrt{2} = 1$ (a rational) but $(1 + \sqrt{2}) + (1 + \sqrt{2}) = 2 + 2\sqrt{2}$ is irrational. The product of two irrational numbers may or may not be irrational. For example, $\sqrt{2}(1 + \sqrt{2}) = \sqrt{2} + 2$ is irrational but $(\sqrt{2})^2 = 2$ is rational.

□

Problem 9

Solution:

Let $k > 1$ be a positive irrational. Note that this is a fair assumption because we show in exercise 11 that $\sqrt{2}$ (which is > 1) is irrational. Also, as in exercise 6, we know $y - x > 0$. Then, by the Archimedean property, there exists an integer n such that $n(y - x) > k$. We can rewrite this as $nx < ny - k$. Now, by exercise 5, we know that there exists a unique integer m such that $nx \leq m < nx + 1$. Hence, $x \leq \frac{m}{n}$ and $m < nx + 1 \implies m < ny - k + 1 \implies m + (k - 1) < ny \implies \frac{m + (k - 1)}{n} < y$. We can combine these two parts to state $x \leq \frac{m}{n} < \frac{m + (k - 1)}{n} < y$, or, more simply, $x < \frac{m}{n} + \frac{k - 1}{n} < y$. Now, we know $n \neq 0$ because otherwise $n(y - x) \not> k$. Hence, $\frac{m}{n}$ is rational. But, by problem 7, $\frac{k - 1}{n}$ is irrational; $k - 1$ is the sum of an irrational and a rational respectively (which is irrational) and $k - 1 \cdot \frac{1}{n}$ is the product of an irrational and a rational respectively (which is again irrational). Thus, $\frac{m}{n} + \frac{k - 1}{n}$ is the sum of a rational and an irrational respectively, which is irrational.

Now, observe that we can repeat the procedure over and over, i.e. we can find an irrational r_1 such that $x < r_1 < \frac{m}{n} + \frac{k - 1}{n} < y$, an irrational r_2 such that $x < r_2 < r_1 < y$, and so on. This gives us a sequence of irrationals $r_1, r_2, \dots$ and from theorem 1.28, we know that there is no upper bound for the set of positive integers. Thus, the existence of one irrational between arbitrary reals implies the existence of infinitely many.

□

Problem 10

Solution:

Lemma: If a, b are integers such that $a < b$, $a + 1 \leq b$.

Proof: Suppose by way of contradiction that $a + 1 > b$. Then, $b - a < 1$. However, since $a < b$, we also have $b - a > 0$. When put together, we get $0 < b - a < 1$. Now, since a, b are both integers $b - a$ is also an integer. But there are no integers > 0 but < 1 . Contradiction!

(a) Suppose that there exists an integer n that is both even and odd. That is, there exist integers m, k such that $n = 2m$ and $n + 1 = 2k$. Hence, $2m + 1 = 2k$. Now, observe that $m < k$ because otherwise $m \geq k \implies 2m \geq 2k \implies 2m + 1 > 2k$, which is a contradiction.

Here, we can invoke the lemma to state $m+1 \leq k$, which implies $2(m+1) = (2m+1)+1 \leq 2k$. But $(2m+1) \leq 2k-1 < 2k$ so we have $2m+1 < 2k$. Contradiction!

(b) Consider an arbitrary integer n . If $\exists m \in \mathbb{Z}$ such that $n = 2m$, then n is even.

Now, we'll show what happens when there isn't. Take the set $S = \{q \in \mathbb{Z}^{\text{even}} : q \leq n\}$.

Detour: we'll quickly show that S has a maximum element. Suppose that this is not the case. Then, $\forall s \in S, s \leq n$, or, in other words, n is an upper bound for the set of even integers. But $\mathbb{Z}^+ \cap \mathbb{Z}^{\text{even}} \subseteq \mathbb{Z}^{\text{even}}$, so $\forall s \in \mathbb{Z}^+ \cap \mathbb{Z}^{\text{even}}, s \in \mathbb{Z}^{\text{even}} \implies s \leq n$, i.e. n is also an upper bound for the set of positive evens. In other words, $\forall k \in \mathbb{Z}^+, 0 < 2k \leq n$. But this means that $0 < 2k \leq n < 2n$ and thereby $\forall k \in \mathbb{Z}^+ k < n$. But this is a contradiction by theorem 1.28.

Since $\max(S)$, exists, $\sup(S) = \max(S)$. Let y be this least upper bound. We will now show that $n - y = n - 2k = 1$ must be true. $y = 2k$ (for some k) since $y \in S$ (and thus even).

Suppose that $n - 2k > 1$. By the lemma, we have $n - 2k \geq 2$, which implies $n \geq 2k + 2 = 2(k + 1)$. Hence, $k + 1 \in S$. However, this is a contradiction because $k = \max(S)$. Now, suppose $n - 2k < 1$. Since we also have $2k < n \implies n - 2k > 0$, we can state $0 < n - 2k < 1$. But, as similarly called out in part (a)'s proof, $n - 2k$ is an integer, but there are no integers > 0 and < 1 . Contradiction again! Hence, by the trichotomy property $n - 2k = 1$.

Now, observe that $n - 2k = 1 \implies n - 2k + 1 = 2 \implies n + 1 = 2(k + 1)$. So n is odd! Thus, we have successfully shown that any integer n can only be even or odd; there are no other options.

(c) Let x, y be arbitrary even integers. Since they're even $x = 2r$ and $y = 2s$ for some integer r, s . Hence, $x + y = 2r + 2s = 2(r + s)$ so the sum of even integers is also even. And $x \cdot y = 2r(2s) = 2(2rs)$ so the product of even integers is also even.

Let a, b be arbitrary odd integers. Now,

$$\begin{aligned} a + b &= ((a + 1) - 1) + ((b + 1) - 1) \\ &= (a + 1) + (b + 1) - 2 \end{aligned}$$

Since sums of even integers are even, there exists $m \in \mathbb{Z}$ such that $(a + 1) + (b + 1) = 2m$.

Hence,

$$\begin{aligned} a + b &= 2m - 2 \\ &= 2(m - 1) \end{aligned}$$

so the sum of odd integers is also even. Now, for the product of odd integers,

$$\begin{aligned} ab &= ((a+1) - 1)((b+1) - 1) \\ &= (a+1)((b+1) - 1) - ((b+1) - 1) \\ &= (a+1)(b+1) - (a+1) - (b+1) + 1 \end{aligned}$$

Since $a+1$ and $b+1$ are even, $\exists p, q \in \mathbb{Z}$ such that $(a+1) = 2p$ and $(b+1) = 2q$. Hence,

$$\begin{aligned} ab &= (2p)(2q) - 2p - 2q + 1 \\ &= 2(p \cdot 2q) - 2p - 2q + 1 \\ &= 2(p \cdot 2q - p - q) + 1 \end{aligned}$$

so $ab + 1 = 2(p \cdot 2q - p - q) + 1 + 1 = 2(p \cdot 2q - p - q + 1)$.

Hence, the product of odd integers is odd.

(d) If n is odd then $n^2 = n \cdot n$ is odd by part (c). Therefore, we can state (by contrapositive) $n^2 \in \mathbb{Z}^{\text{even}} \implies n \in \mathbb{Z}^{\text{even}}$. Now, if we have $a^2 = 2b^2$, then a^2 is even (i.e. there exists an m such that $a^2 = 2m$). Hence, by the previous theorem in this part, a is also even. Let $a = 2k$. Then, $a^2 = (2k)^2 = (2k)(2k) = 2(2k^2) = 2b^2$. This gives us $2k^2 = b^2$ so b^2 is also even. Finally, by the previous theorem in this part, b is also even.

(e) We'll prove it by contradiction. Let $q = \frac{a}{b} \in \mathbb{Q}$ be a rational where a, b are both even integers ($b \neq 0$). Then, observe that $\frac{a}{b} = \frac{2u}{2v}$ (for some integer u, v) $= (\frac{2}{2}) \frac{u}{v} = (1) \frac{u}{v} = \frac{u}{v}$. As long as u, v are both even, we can keep repeating the same procedure until either the numerator or denominator is odd. And at that point we'll have shown a contradiction because either a or b will be odd.

Note that we can't yet rigorously show that this "repeatedly extracting 2 from the number" procedure will stop at some point (which is required if we want to use this argument), because we haven't yet developed the machinery of induction. $\square$

Problem 11

Solution:

Proof by contradiction: suppose that there exists a $q = \frac{a}{b} \in \mathbb{Q}$ (a, b are integers, at least one of which is odd) such that $q^2 = 2$. Then, $q^2 = (\frac{a}{b})^2 = (ab^{-1})(ab^{-1}) = a^2(b^{-1}b^{-1}) = \frac{a^2}{b \cdot b} = \frac{a^2}{b^2} = 2$, which implies $a^2 = 2b^2$. But from part (d), we know that this implies that both a, b are even. Contradiction!

$\square$

Problem 12

Solution:

We'll start by showing that the rationals satisfy the Archimedean property. That is, given $x = \frac{a}{b} \in \mathbb{Q}$ (a, b are integers with $b \neq 0$) where $x > 0$ and arbitrary $y = \frac{c}{d} \in \mathbb{Q}$ (c, d are integers with $d \neq 0$), there exists an integer n such that $nx > y$.

Proof: Observe that $\left(\frac{c}{d}\right)\left(\frac{b}{a}\right) = \frac{bc}{ad}$. Also, $a \neq 0$ because otherwise, $a = bx = 0$ which implies that either $b = 0$ or $x = 0$ (and either would give a contradiction). Now, $a \neq 0$ and $d \neq 0$ means $ad \neq 0$. Hence, $\frac{bc}{ad}$ is rational (and thereby also real). By theorem 1.29, there exists a positive $n > \frac{bc}{ad}$, which implies that $n\left(\frac{a}{b}\right) > \frac{c}{d}$. This can be rewritten as $nx > y$.

We are disproving the following statement: Every nonempty set S of rational numbers ($S \subseteq \mathbb{Q}$) which is bounded above by a rational number has a supremum in $\mathbb{Q}$ (that is, there is a rational number b such that $b = \sup S$). We will do so by example.

Consider $S = \{q \in \mathbb{Q} : q^2 < 2\}$. S is non-empty because it contains at least 0 ($0^2 = 0 \cdot 0 = 0 < 2$). It is also bounded above by 2 because $q^2 < 2 \implies q < 2$ (otherwise, we would have $q \geq 2 \implies q^2 \geq 2^2 = 4 > 2$ which contradicts our set construction). Hence, the theorem applies.

We want to show $\sup S = k$, where k is the irrational (per problem 11) such that $k^2 = 2$.

To start, observe that $q < k$ because otherwise $q^2 \geq k^2 = 2$ (which contradicts our set construction). Therefore k is an upper bound for S . We will now show that if u is an upper bound for S , $u \geq k$. We'll use a proof by contradiction.

Let u be an upper bound for S where $u < k$. Note that $0 \leq u$ because $0 \in S$ and u is an upper bound. From problem 6, we know that there exists a rational r such that $u < r < k$. But observe that $r \geq 0 \wedge r < k \implies r^2 < k^2 = 2$. Then, since r is rational, $r \in S$. This is a contradiction since u is supposed to be an upper bound for S , but $u < r$.

Since any upper bound for S is $\geq k$, $k = \sup S$. And since k is a known irrational (from problem 11), we've successfully disproved the statement that the supremum of every non-empty + bounded-above set of rationals is a rational. $\square$